

HW7

$$1. (a) x(t) = \sum_{-\infty}^{\infty} a_k e^{j\omega_0 k t} \quad x(t-t_0) = \sum_{-\infty}^{\infty} a_k e^{-j\omega_0 k t_0} e^{j\omega_0 k t}$$

$$x(t+t_0) = \sum_{-\infty}^{\infty} a_k e^{j\omega_0 k t_0} e^{j\omega_0 k t}$$

$$x(t-t_0) + x(t+t_0) = \sum_{-\infty}^{\infty} a_k (e^{j\omega_0 k t_0} + e^{-j\omega_0 k t_0}) e^{j\omega_0 k t}$$

$$= \sum_{-\infty}^{\infty} 2a_k \cos \frac{2\pi}{T} k t_0 e^{j\omega_0 k t}$$

$$\therefore b_k = 2a_k \cos \frac{2\pi}{T} k t_0$$

$$(b) x(t) = \sum_{-\infty}^{\infty} a_k e^{j\omega_0 k t} \quad x(-t) = \sum_{-\infty}^{\infty} a_k e^{-j\omega_0 k t} = \sum_{-\infty}^{\infty} a_{-k} e^{j\omega_0 k t}$$

$$\text{Ev } x(t) = \frac{x(t) + x(-t)}{2} = \frac{1}{2} \sum_{-\infty}^{\infty} (a_k + a_{-k}) e^{j\omega_0 k t}$$

$$\therefore b_k = \frac{1}{2} (a_k + a_{-k})$$

$$(c) x(t) = \sum_{-\infty}^{\infty} a_k e^{j\omega_0 k t} \quad x^*(t) = \sum_{-\infty}^{\infty} a_k^* e^{j\omega_0 k t}$$

$$\text{Re } x(t) = \frac{1}{2} (x(t) + x^*(t)) = \sum_{-\infty}^{\infty} \frac{1}{2} (a_k + a_k^*) e^{j\omega_0 k t}$$

$$\therefore b_k = \frac{1}{2} (a_k + a_k^*)$$

$$(d) x(t) = \sum_{-\infty}^{\infty} a_k e^{j\omega_0 k t} \quad \frac{d^2 x(t)}{dt^2} = \sum_{-\infty}^{\infty} (j\omega_0 k)^2 a_k e^{j\omega_0 k t}$$

$$\therefore b_k = -(\omega_0 k)^2 a_k$$

$$2. T=2 \Rightarrow \omega_0 = \frac{2\pi}{T} = \pi$$

$$\int_{-0.5}^{1.5} x(t) dt = 2+1=3 \quad a_0 = \frac{1}{2} \int_{-0.5}^1 x(t) dt = \frac{3}{2}$$

$$\therefore a_k = a_{k+2} \quad \therefore \text{when } k \text{ is even } a_k = \frac{3}{2}$$

$$\therefore a_k \text{ is periodic } \therefore x(t) \text{ is a impulse train}$$

$$a_k = \frac{1}{T} = \frac{1}{2} \quad \therefore \text{when } k \text{ is odd } a_k = \frac{1}{2}$$

$$3. (a) \textcircled{1} \text{ let } a_k = a + bj \quad a_{-k} = c + dj$$

$$a_k e^{j\omega_0 k t} + a_{-k} e^{-j\omega_0 k t}$$

$$= (a + bj)(\cos \omega_0 k t + j \sin \omega_0 k t) + (c + dj)(\cos \omega_0 k t - j \sin \omega_0 k t)$$

$$= (a + c) \cos \omega_0 k t + (-b + d) \sin \omega_0 k t + (b + d) j \cos \omega_0 k t$$

$$+ (a - c) j \sin \omega_0 k t$$

$$\therefore x(t) \text{ is real valued } \therefore a = c \quad b = -d \quad a_k = a_{-k}^*$$

$$\textcircled{2} a_k = \frac{1}{T} \int_T x(t) e^{-j\omega_0 k t} dt \quad a_{-k} = \frac{1}{T} \int_T x(t) e^{j\omega_0 k t} dt$$

$$a_k^* = \frac{1}{T} \int_T x(t) e^{j\omega_0 k t} dt = a_{-k} \quad \therefore a_k = a_{-k}^*$$

$$(b) x(t) = x(-t) \quad x(t) \rightarrow a_k \quad x(-t) = a_{-k} \quad \therefore a_k = a_{-k}$$

$$\therefore a_k = a_{-k}^* \quad \therefore a_{-k} = a_k^* \quad \therefore a_k \text{ is real and even}$$

$$(c) x(t) = -x(-t) \quad \therefore a_k = -a_{-k}$$

$$\therefore a_k = a_{-k}^* \quad \therefore -a_{-k} = a_{-k}^* \quad \therefore a_k \text{ is pure imaginary}$$

$$\text{and odd } \therefore x(t) \text{ is odd } \therefore \int_T x(t) dt = 0 \quad \therefore a_0 = 0$$

$$4. (1) z(t) = \sum_{m=-\infty}^{\infty} a_m e^{j\omega_0 m t} \sum_{n=-\infty}^{\infty} b_n e^{j\omega_0 n t}$$

$$= \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} a_n b_{k-n} e^{j\omega_0 (m+n)t} \quad m+n=k$$

$$\therefore c_k = \sum_{n=-\infty}^{\infty} a_n b_{k-n}$$

$$(2) z(t) = \frac{1}{T} \int_T \sum_{n=-\infty}^{\infty} a_n e^{j\omega_0 n \tau} \sum_{m=-\infty}^{\infty} b_m e^{j\omega_0 m (t-\tau)} d\tau$$

$$= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} a_n e^{j\omega_0 n t} b_m \frac{1}{T} \int_T e^{j\omega_0 (n-m)\tau} d\tau$$

$$\therefore n=m \quad \sum_{n=-\infty}^{\infty} a_n b_n e^{j\omega_0 n t}$$

$$\therefore c_k = a_k \cdot b_k$$